Week 10 Report: Perform Task Action

Autonomous Warehouse Inventory and Item Delivery Robot

Course	TTTC2343
Week	10
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Project Title	Autonomous Warehouse Inventory and Item Delivery Robot Simulation

Week 10 Progress Report: Perform Task Action

Project Title: Customized TIAGo / TurtleBot3 Simulation Development **Topic:** Task Execution (QR Code Scanning & Simulated Item Delivery)

1. Introduction

The objective for Week 10 is to implement and demonstrate a specific task action within the warehouse simulation environment, specifically "QR code scanning OR Pick-and-place operation".

In our custom Gazebo warehouse simulation, the TurtleBot3 robot has been programmed to perform an autonomous inventory verification and item delivery task. This meets the criteria by simulating marker-based inventory scanning (analogous to QR code scanning) and drop-off confirmation.

2. Simulation Environment Setup

The customized AWS Small Warehouse environment has been configured with designated inventory areas, charging stations, and packing/drop-off stations.

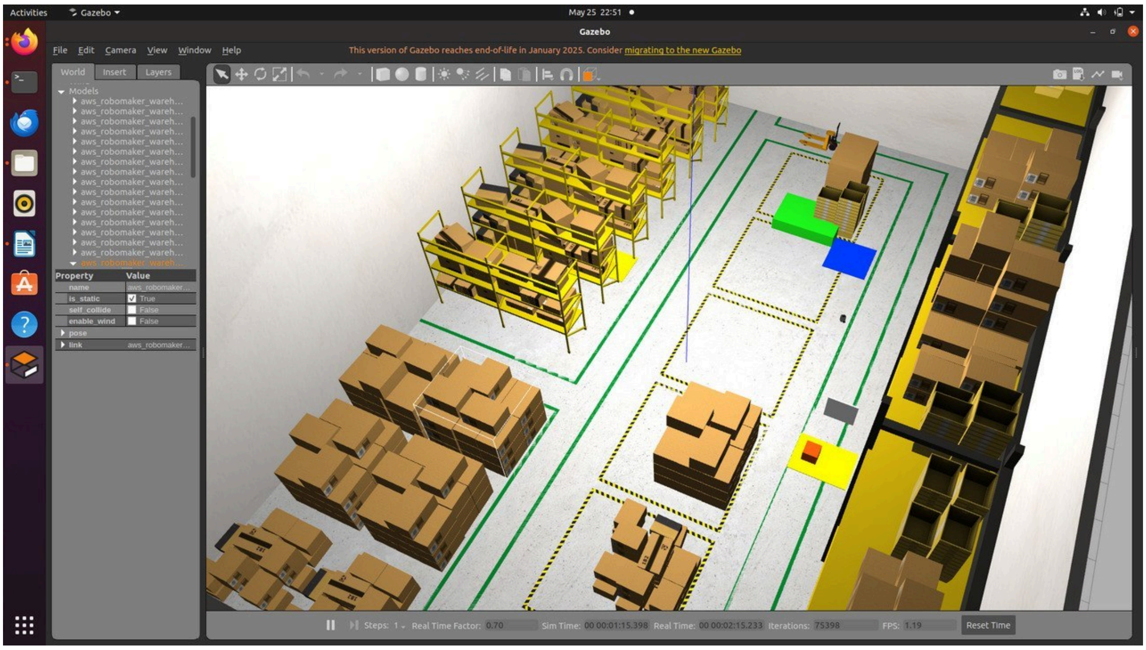

Figure 1: Gazebo Simulation Environment showing the warehouse layout and the robot.

3. Task Execution: Marker-Based Inventory Verification

During the mission, the robot navigates autonomously to designated inventory shelves. Upon arrival, it performs a simulated marker scan to verify the inventory label against the system's records.

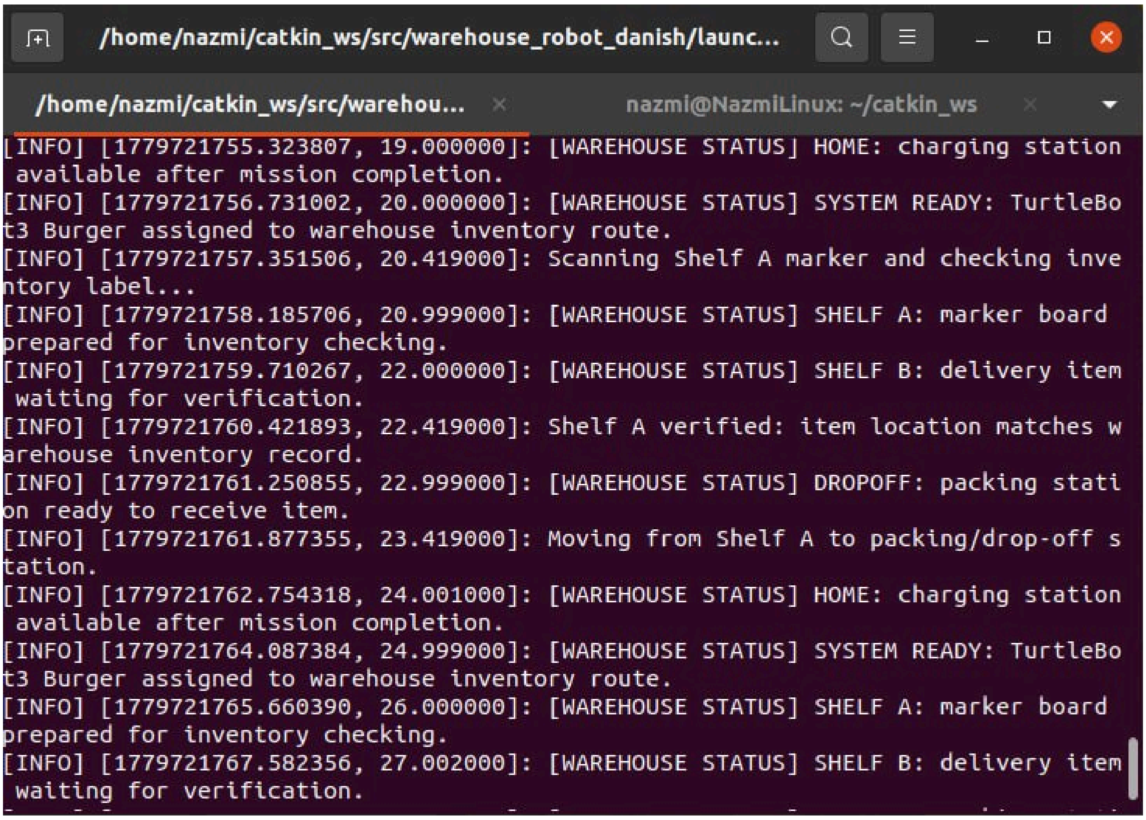

Figure 2: Terminal output confirming the robot is "Scanning Shelf A marker and checking inventory label..." which satisfies the scanning task requirement.

4. Item Delivery and Mission Completion

After successful verification at the shelf, the robot proceeds to the packing/drop-off station. It simulates delivering the verified item to the packing area.

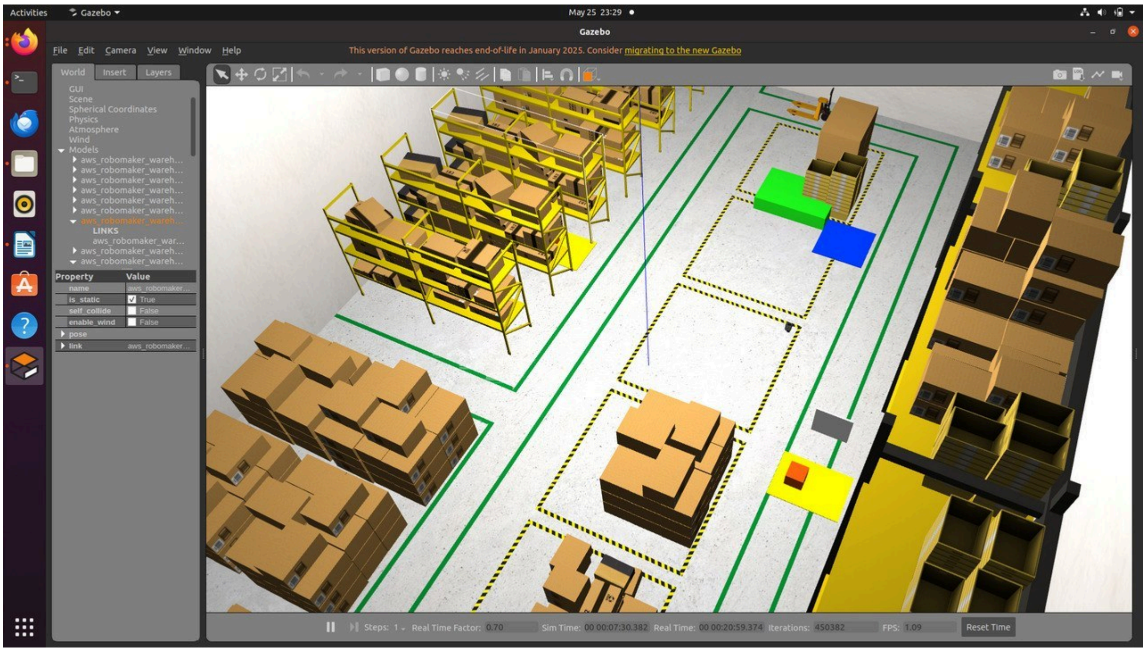

Figure 3: The robot is positioned at the drop-off station after completing the simulated delivery.

The mission controller logs the successful delivery and signals that the robot is ready to return to standby or execute the next task, completely automating the end-to-end mission process.

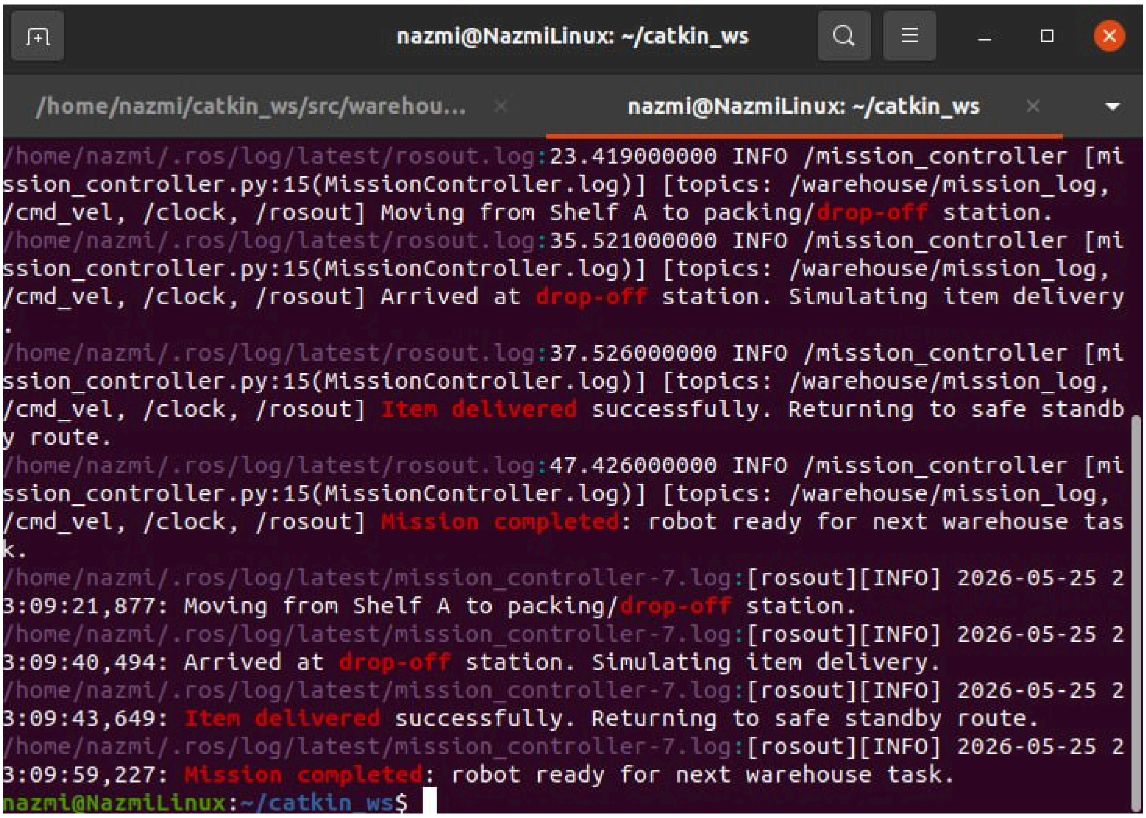

Figure 4: Terminal logs showing "Item delivered successfully" and "Mission completed".

5. Conclusion

The simulated actions shown in the logs and environment successfully fulfill the Week 10 requirements. The robot correctly executes its assigned tasks (scanning and delivery verification) within the designated Gazebo warehouse zones.

6. Video Demonstration

A full video demonstration of the task execution can be viewed here: Week 10 Task Demo Video